1) a)

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (y, -x)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{y^2}{3} + \frac{x^2}{4} = 1 \therefore \frac{x^2}{4} + \frac{y^2}{3} = 1$

Enlargement of scale factor 2: $(x, y) \longmapsto \left(\frac{x}{2}, \frac{y}{2}\right)$

$\frac{x^2}{4} + \frac{y^2}{3} = 1 \longmapsto \frac{\left(\frac{x}{2}\right)^2}{4} + \frac{\left(\frac{y}{2}\right)^2}{3} = 1 \therefore \frac{x^2}{16} + \frac{y^2}{12} = 1$

1) b)

Reflection in the x-axis: $(x, y) \longmapsto (x, -y)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{x^2}{3} + \frac{y^2}{4} = 1 \therefore \frac{x^2}{3} + \frac{y^2}{4} = 1$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (y, -x)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{y^2}{3} + \frac{x^2}{4} = 1 \therefore \frac{x^2}{4} + \frac{y^2}{3} = 1$

1) c)

Translation by vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$: $(x, y) \longmapsto (x - 1, y + 2)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{(x-1)^2}{3} + \frac{(y+2)^2}{4} = 1$

Reflection in the $y = -x$: $(x, y) \longmapsto (-y, -x)$

$\frac{(x-1)^2}{3} + \frac{(y+2)^2}{4} = 1 \longmapsto \frac{(-y-1)^2}{3} + \frac{(-x+2)^2}{4} = 1 \therefore \frac{(x-2)^2}{4} + \frac{(y+1)^2}{3} = 1$

1) d)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (-y, x)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{y^2}{3} + \frac{x^2}{4} = 1$

Enlargement of scale factor $\frac{1}{2}$: $(x, y) \longmapsto (2x, 2y)$

$\frac{y^2}{3} + \frac{x^2}{4} = 1 \longmapsto \frac{(2y)^2}{3} + \frac{(2x)^2}{4} = 1 \therefore x^2 + \frac{4y^2}{3} = 1$

1) e)

Stretch parallel to x-axis, scale factor 3: $(x, y) \longmapsto \left(\frac{x}{3}, y\right)$

$\frac{x^2}{3} + \frac{y^2}{4} = 1 \longmapsto \frac{\left(\frac{x}{3}\right)^2}{3} + \frac{y^2}{4} = 1 \therefore \frac{x^2}{27} + \frac{y^2}{4} = 1$

Translation by vector $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$: $(x, y) \longmapsto (x + 1, y - 1)$

$\frac{x^2}{27} + \frac{y^2}{4} = 1 \longmapsto \frac{(x+1)^2}{27} + \frac{(y-1)^2}{4} = 1$

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (-y, x)$

$\frac{(x+1)^2}{27} + \frac{(y-1)^2}{4} = 1 \longmapsto \frac{(-y+1)^2}{27} + \frac{(x-1)^2}{4} = 1 \therefore \frac{(x-1)^2}{4} + \frac{(y-1)^2}{27} = 1$

2) a)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (-y, x)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{y^2}{6} - \frac{x^2}{3} = 1$

Stretch parallel to y-axis, scale factor 3: $(x, y) \longmapsto \left(x, \frac{y}{4}\right)$

$\frac{y^2}{6} - \frac{x^2}{3} = 1 \longmapsto \frac{\left(\frac{y}{4}\right)^2}{6} - \frac{x^2}{3} = 1 \therefore \frac{y^2}{96} - \frac{x^2}{3} = 1$

2) b)

Reflection in the y-axis: $(x, y) \longmapsto (-x, y)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{x^2}{6} - \frac{y^2}{3} = 1$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (y, -x)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{y^2}{6} - \frac{(-x)^2}{3} = 1 \therefore \frac{y^2}{6} - \frac{x^2}{3} = 1$

2) c)

Translation by vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$: $(x, y) \longmapsto (x + 1, y - 3)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{(x+1)^2}{6} - \frac{(y-3)^2}{3} = 1$

Reflection in the line $y = x$: $(x, y) \longmapsto (y, x)$

$\frac{(x+1)^2}{6} - \frac{(y-3)^2}{3} = 1 \longmapsto \frac{(y+1)^2}{6} - \frac{(x-3)^2}{3} = 1$

2) d)

Stretch parallel to x-axis, scale factor $\frac{1}{3}$: $(x, y) \longmapsto (3x, y)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{(3x)^2}{6} - \frac{y^2}{3} = 1$

Reflection in the $y = -x$: $(x, y) \longmapsto (-y, -x)$

$\frac{3x^2}{2} - \frac{y^2}{3} = 1 \longmapsto \frac{3y^2}{2} - \frac{x^2}{3} = 1$

2) e)

Enlargement of scale factor 2: $(x, y) \longmapsto \left(\frac{x}{2}, \frac{y}{2}\right)$

$\frac{x^2}{6} - \frac{y^2}{3} = 1 \longmapsto \frac{\left(\frac{x}{2}\right)^2}{6} - \frac{\left(\frac{y}{2}\right)^2}{3} = 1 \therefore \frac{x^2}{24} - \frac{y^2}{12} = 1$

Translation by vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $(x, y) \longmapsto (x - 1, y + 1)$

$\frac{x^2}{24} - \frac{y^2}{12} = 1 \longmapsto \frac{(x-1)^2}{24} - \frac{(y+1)^2}{12} = 1$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (y, -x)$

$\frac{(x-1)^2}{24} - \frac{(y+1)^2}{12} = 1 \longmapsto \frac{(y-1)^2}{24} - \frac{(-x+1)^2}{12} = 1 \therefore \frac{(y-1)^2}{24} - \frac{(x-1)^2}{12} = 1$

3) a)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (-y, x)$

$xy = 16 \longmapsto -xy = 16 \Rightarrow xy = -16$

Enlargement of scale factor 3: $(x, y) \longmapsto \left(\frac{x}{3}, \frac{y}{3}\right)$

$xy = -16 \longmapsto \left(\frac{x}{3}\right)\left(\frac{y}{3}\right) = -16 \therefore xy = -144$

3) b)

Reflection in the x-axis: $(x, y) \longmapsto (x, -y)$

$xy = 16 \longmapsto -xy = 16 \Rightarrow xy = -16$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (y, -x)$

$xy = -16 \longmapsto xy = 16$

3) c)

Translation by vector $\begin{pmatrix}1\\2\end{pmatrix}$: $(x, y) \longmapsto (x-1, y-2)$

$xy = 16 \longmapsto (x-1)(y-2) = 16$

Reflection in the $y = -x$: $(x, y) \longmapsto (-y, -x)$

$(x-1)(y-2) = 16 \longmapsto (-y-1)(-x-2) = 16 \therefore (y+1)(x+2) = 16$

3) d)

Stretch parallel to x-axis, scale factor 2: $(x, y) \longmapsto \left(\frac{x}{2}, y\right)$

$xy = 16 \longmapsto \left(\frac{x}{2}\right)y = 16 \Rightarrow xy = 32$

Translation by vector $\begin{pmatrix}-2\\4\end{pmatrix}$: $(x, y) \longmapsto (x+2, y-4)$

$xy = 32 \longmapsto (x+2)(y-4) = 32$

3) e)

Enlargement of scale factor 3: $(x, y) \longmapsto \left(\frac{x}{2}, \frac{y}{2}\right)$

$xy = 16 \longmapsto \left(\frac{x}{2}\right)\left(\frac{y}{2}\right) = 16 \Rightarrow xy = 64$

Translation by vector $\begin{pmatrix}-1\\4\end{pmatrix}$: $(x, y) \longmapsto (x+1, y-4)$

$xy = 64 \longmapsto (x+1)(y-4) = 64$

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \longmapsto (-y, x)$

$(x+1)(y-4) = 64 \longmapsto (-y+1)(x-4) = 64 \therefore (y-1)(x-4) = -64$

4) a)

Stretch parallel to y-axis, scale factor 2: $(x, y) \longmapsto \left(x, \frac{y}{2}\right)$

$\frac{x^2}{5} + \frac{y^2}{12} = 1 \longmapsto \frac{x^2}{5} + \frac{\left(\frac{y}{2}\right)^2}{12} = 1 \Rightarrow \frac{x^2}{5} + \frac{y^2}{48} = 1$

Translation by vector $\begin{pmatrix}3\\-4\end{pmatrix}$: $(x, y) \longmapsto (x-3, y+4)$

$\frac{x^2}{5} + \frac{y^2}{48} = 1 \longmapsto \frac{(x-3)^2}{5} + \frac{(y+4)^2}{48} = 1$

Rotate by π radians anti-clockwise: $(x, y) \longmapsto (-x, -y)$

$\frac{(x-3)^2}{5} + \frac{(y+4)^2}{48} = 1 \longmapsto \frac{(-x-3)^2}{5} + \frac{(-y+4)^2}{48} = 1 \therefore \frac{(x+3)^2}{5} + \frac{(y-4)^2}{48} = 1$

4) b)

Stretch parallel to y-axis, scale factor 2: $(x, y) \longmapsto \left(x, \frac{y}{2}\right)$

$\frac{x^2}{4} - \frac{y^2}{6} = 1 \longmapsto \frac{x^2}{4} - \frac{\left(\frac{y}{2}\right)^2}{6} = 1 \Rightarrow \frac{x^2}{4} - \frac{y^2}{24} = 1$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \longmapsto (x - 3, y + 4)$

$\frac{x^2}{4} - \frac{y^2}{24} = 1 \longmapsto \frac{(x-3)^2}{4} - \frac{(y+4)^2}{24} = 1$

Rotate by π radians anti-clockwise: $(x, y) \longmapsto (-x, -y)$

$\frac{(x-3)^2}{4} - \frac{(y+4)^2}{24} = 1 \longmapsto \frac{(-x-3)^2}{4} - \frac{(-y+4)^2}{24} = 1 \therefore \frac{(x+3)^2}{4} - \frac{(y-4)^2}{24} = 1$

4) c)

Stretch parallel to y-axis, scale factor 2: $(x, y) \longmapsto \left(x, \frac{y}{2}\right)$

$xy = 11 \longmapsto x\left(\frac{y}{2}\right) = 11 \Rightarrow xy = 22$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \longmapsto (x - 3, y + 4)$

$xy = 22 \longmapsto (x - 3)(y + 4) = 22$

Rotate by π radians anti-clockwise: $(x, y) \longmapsto (-x, -y)$

$(x - 3)(y + 4) = 22 \longmapsto (-x - 3)(-y + 4) = 22 \therefore (x + 3)(y - 4) = 22$

4) d)

Stretch parallel to y-axis, scale factor 2: $(x, y) \longmapsto \left(x, \frac{y}{2}\right)$

$x^2 - y^2 = 9 \longmapsto x^2 - \left(\frac{y}{2}\right)^2 = 9 \Rightarrow x^2 - \frac{y^2}{4} = 9$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \longmapsto (x - 3, y + 4)$

$x^2 - \frac{y^2}{4} = 9 \longmapsto (x - 3)^2 - \frac{(y+4)^2}{4} = 9$

Rotate by π radians anti-clockwise: $(x, y) \longmapsto (-x, -y)$

$(x - 3)^2 - \frac{(y+4)^2}{4} = 9 \longmapsto (-x - 3)^2 - \frac{(-y+4)^2}{4} = 9 \therefore (x + 3)^2 - \frac{(y-4)^2}{4} = 9$

5) a)

$x^2 + 9y^2 = 1 \longmapsto \frac{x^2}{36} + \frac{y^2}{4} = 1 \longmapsto \frac{(x-2)^2}{36} + \frac{y^2}{4} = 1$

Enlargement of scale factor 6 about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

Or

$x^2 + 9y^2 = 1 \longmapsto \left(x - \frac{1}{3}\right)^2 + 9y^2 = 1 \longmapsto \frac{(x-2)^2}{36} + \frac{y^2}{4} = 1$

Translation by vector $\begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix}$ followed by an enlargement of scale factor 6 about $(0, 0)$

5) b)

$\frac{x^2}{9} - \frac{y^2}{4} = 1 \longmapsto \frac{x^2}{9} - \frac{y^2}{16} = 1 \longmapsto \frac{(x-3)^2}{9} - \frac{y^2}{16} = 1$

Stretch parallel to y-axis, scale factor 2 and translation by vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ (any order)

5) c)

$xy = 1 \longmapsto (x-2)y = 1 \longmapsto (x-2)y = 3$

Translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ followed by a stretch parallel to y-axis, scale factor 3 (any order)

Or

$xy = 1 \longmapsto \left(\frac{x}{3}\right)y = 1 \longmapsto (x-2)y = 3$

Stretch parallel to x-axis, scale factor 3 followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

Or

$xy = 1 \longmapsto xy = 3 \longmapsto (x-2)y = 3$

Enlargement of scale factor $\sqrt{3}$ about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

5) d)

$\frac{x^2}{16}+\frac{y^2}{9}=1 \mapsto \frac{x^2}{16}+\frac{y^2}{36}=1 \mapsto \frac{(x-1)^2}{16}+\frac{(y+3)^2}{36}=1$

Stretch parallel to y-axis, scale factor 2 followed by a translation by vector $\begin{pmatrix}1\\-3\end{pmatrix}$

Or

$\frac{x^2}{16}+\frac{y^2}{9}=1 \mapsto \frac{(x-1)^2}{16}+\frac{\left(y+\frac{3}{2}\right)^2}{9}=1 \mapsto \frac{(x-1)^2}{16}+\frac{(y+3)^2}{36}=1$

Translation by vector $\begin{pmatrix}1\\-\frac{3}{2}\end{pmatrix}$ followed by a stretch parallel to y-axis, scale factor 2

5) e)

$\frac{x^2}{4}-\frac{y^2}{9}=1 \mapsto \frac{x^2}{8}-\frac{y^2}{18} \mapsto \frac{(x-2)^2}{8}-\frac{y^2}{18}=1 \Rightarrow \frac{(2-x)^2}{8}-\frac{y^2}{18}=1$

Enlargement of scale factor $\sqrt{2}$ about $(0,0)$ followed by a translation by vector $\begin{pmatrix}2\\0\end{pmatrix}$

Or

$\frac{x^2}{4}-\frac{y^2}{9}=1 \mapsto \frac{\left(x-\sqrt{2}\right)^2}{4}-\frac{y^2}{9}=1 \mapsto \frac{(x-2)^2}{8}-\frac{y^2}{18}=1 \Rightarrow \frac{(2-x)^2}{8}-\frac{y^2}{18}=1$

Translation by vector $\begin{pmatrix}\sqrt{2}\\0\end{pmatrix}$ followed by an enlargement of scale factor $\sqrt{2}$ about $(0,0)$

5) f)

$\frac{x^2}{25}+\frac{y^2}{4}=1 \mapsto \frac{x^2}{4}+\frac{y^2}{25}=1 \mapsto \frac{(x+2)^2}{4}+\frac{(y-1)^2}{25}=1$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise OR Rotate by $\frac{3\pi}{2}$ radians anti-clockwise OR reflection in $y=x$ OR reflection in $y=-x$ AND a translation by vector $\begin{pmatrix}-2\\1\end{pmatrix}$ (any order)

5) g)

$xy = 1 \longmapsto xy = 4 \longmapsto (x+1)(y-2) = 4$

Enlargement of scale factor 2 about $(0,0)$ followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Or

$xy = 1 \longmapsto \left(\frac{x}{4}\right) y = 1 \longmapsto (x+1)(y-2) = 4$

Stretch parallel to x-axis, scale factor 4 followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Or

$xy = 1 \longmapsto x\left(\frac{y}{4}\right) = 1 \longmapsto (x+1)(y-2) = 4$

Stretch parallel to y-axis, scale factor 4 followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Or

$xy = 1 \longmapsto \left(x + \frac{1}{2}\right)(y-1) = 1 \longmapsto \left(\frac{1}{2}x + \frac{1}{2}\right)\left(\frac{1}{2}y - 1\right) = 1 \Rightarrow (x+1)(y-2) = 4$

Translation by vector $\begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}$ followed by an enlargement of scale factor 2 about $(0,0)$

5) h)

$9(y-1)^2 - 4x^2 = 144 \Rightarrow \frac{9(y-1)^2}{144} - \frac{4x^2}{144} = 1 \Rightarrow \frac{(y-1)^2}{16} - \frac{x^2}{36} = 1$

$\frac{y^2}{16} - \frac{x^2}{9} = 1 \longmapsto \frac{y^2}{16} - \frac{x^2}{36} = 1 \longmapsto \frac{(y-1)^2}{16} - \frac{x^2}{36} = 1$

Stretch parallel to x-axis, scale factor 2 and a translation by vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (any order)

5) i)

$x^2 + 5(y^2 - 4y) = -5 \Rightarrow x^2 + 5(y-2)^2 - 20 = -5 \Rightarrow x^2 + 5(y-2)^2 = 15$

$\Rightarrow \frac{x^2}{15} + \frac{(y-2)^2}{3} = 1$

$\frac{x^2}{5} + \frac{y^2}{3} = 1 \longmapsto \frac{x^2}{5} + \frac{(y-2)^2}{3} = 1 \longmapsto \frac{x^2}{15} + \frac{(y-2)^2}{3} = 1$

Stretch parallel to x-axis, scale factor $\sqrt{3}$ followed by a translation by vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ (any order)

5) j)

$x^2 = 5(y^2 + 2x) \Rightarrow x^2 - 10x - 5y^2 = 0 \Rightarrow (x-5)^2 - 25 - 5y^2 = 0$

$\Rightarrow (x-5)^2 - 5y^2 = 25 \Rightarrow \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1$

$x^2 - \frac{y^2}{5} = 1 \mapsto \frac{x^2}{25} - \frac{y^2}{5} = 1 \mapsto \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1$

Stretch parallel to x-axis, scale factor 5 followed by a translation by vector $\begin{pmatrix}5\\0\end{pmatrix}$

Or

$x^2 - \frac{y^2}{5} = 1 \mapsto (x-1)^2 - \frac{y^2}{5} = 1 \mapsto \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1$

Translation by vector $\begin{pmatrix}1\\0\end{pmatrix}$ followed by a stretch parallel to x-axis, scale factor 5

6) a)

$2(x-1)^2 + (y+3)^2 = 16 \Rightarrow \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1$

$\frac{x^2}{2} + \frac{y^2}{4} = 1 \mapsto \frac{x^2}{8} + \frac{y^2}{16} = 1 \mapsto \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1$

Enlargement of scale factor 2 about $(0,0)$ followed by a translation by vector $\begin{pmatrix}1\\-3\end{pmatrix}$

Or

$\frac{x^2}{2} + \frac{y^2}{4} = 1 \mapsto \frac{\left(x-\frac{1}{2}\right)^2}{2} + \frac{\left(y+\frac{3}{2}\right)^2}{4} = 1 \mapsto \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1$

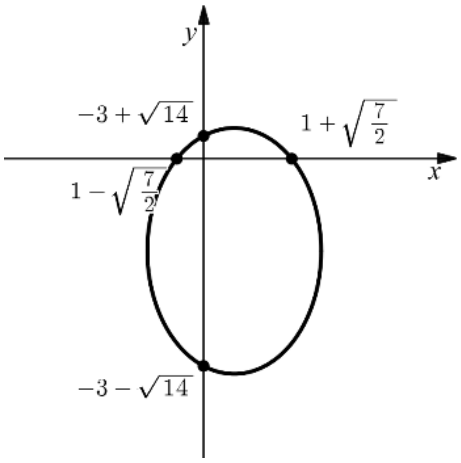
Translation by vector $\begin{pmatrix}-\frac{1}{2}\\\frac{3}{2}\end{pmatrix}$ followed by an enlargement of scale factor 2 about $(0,0)$

6) b)

When $x = 0$, $2(0-1)^2 + (y+3)^2 = 16 \Rightarrow y = -3 \pm \sqrt{14}$

When $y = 0$, $2(x-1)^2 + 3^2 = 16 \Rightarrow y = 1 \pm \sqrt{\frac{7}{2}}$

6) c)



7) a)

Hyperbola

7) b)

Asymptotes are $y = \frac{4}{3}x$ and $y = -\frac{4}{3}x$

7) c)

Translation by vector $\begin{pmatrix}4\\1\end{pmatrix}$: $(x, y) \longmapsto (x-4, y-1)$

$y = \pm\frac{4}{3}x \longmapsto y - 1 = \pm\frac{4}{3}(x-4)$

Reflection in the line $y = x$: $(x, y) \longmapsto (y, x)$

$y - 1 = \pm\frac{4}{3}(x-4) \longmapsto x - 1 = \pm\frac{4}{3}(y-4)$

So $x - 1 = \frac{4}{3}(y-4) \Rightarrow y = \frac{3}{4}x + \frac{13}{4}$ or $x - 1 = -\frac{4}{3}(y-4) \Rightarrow y = -\frac{3}{4}x + \frac{19}{4}$

$\therefore y = \frac{3}{4}x + \frac{13}{4}$ or $y = -\frac{3}{4}x + \frac{19}{4}$

8) a)

Rotation by $\frac{\pi}{2}$ anti-clockwise followed by an enlargement about the origin scale factor 2 (although order doesn't matter).

8) b)

$\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2y \\ 2x \end{pmatrix}$

So that $X = -2y$ and $Y = 2x \Rightarrow x = \frac{Y}{2}, y = -\frac{X}{2}$

So $x^2 + 4y^2 = 1 \longmapsto \left(\frac{Y}{2}\right)^2 + 4\left(-\frac{X}{2}\right)^2 = 1 \Rightarrow \frac{Y^2}{4} + X^2 = 1$

$\therefore x^2 + \frac{y^2}{4} = 1$

9) a)

Enlargement about the origin scale factor 4, followed by a reflection across the x-axis.

9) b)

$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & -4 \end{pmatrix}$

$\begin{pmatrix} 4 & 0 \\ 0 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4x \\ -4y \end{pmatrix}$

So that $X = 4x$ and $Y = -4y \Rightarrow x = \frac{X}{4}, y = -\frac{Y}{4}$

So $x^2 - y^2 = 4 \longmapsto \left(\frac{X}{4}\right)^2 - \left(-\frac{Y}{4}\right)^2 = 4 \Rightarrow \frac{X^2}{16} - \frac{Y^2}{16} = 4$

$\therefore x^2 - y^2 = 64$

10) a)

$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2}$ $\therefore y = \frac{1}{2}x$ or $y = -\frac{1}{2}x$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ so $y = -3x$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$ so $y = x$

$\therefore y = -3x$ and $y = x$

10) b)

$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2}$ $\therefore y = \frac{1}{2}x$ or $y = -\frac{1}{2}x$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix}2\\1\end{pmatrix} \Rightarrow \begin{pmatrix}2 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}5\\2\end{pmatrix}$ so $y = \frac{2}{5}x$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix}-2\\1\end{pmatrix} \Rightarrow \begin{pmatrix}2 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}-3\\-2\end{pmatrix}$ so $y = \frac{2}{3}x$

$\therefore y = \frac{2}{5}x$ and $y = \frac{2}{3}x$

10) c)

$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2}$ $\therefore y = \frac{1}{2}x$ or $y = -\frac{1}{2}x$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix}2\\1\end{pmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}3 & -1\\3 & 1\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}\frac{5}{\sqrt{2}}\\\frac{7}{\sqrt{2}}\end{pmatrix}$ so $y = \frac{7}{5}x$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix}-2\\1\end{pmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}3 & -1\\3 & 1\end{pmatrix}\begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}-\frac{7}{\sqrt{2}}\\-\frac{5}{\sqrt{2}}\end{pmatrix}$ so $y = \frac{5}{7}x$

$\therefore y = \frac{7}{5}x$ and $y = \frac{5}{7}x$

10) d)

$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2}$ $\therefore y = \frac{1}{2}x$ or $y = -\frac{1}{2}x$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix}2\\1\end{pmatrix} \Rightarrow \begin{pmatrix}1 & 2\\0 & -1\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}4\\-1\end{pmatrix}$ so $y = -\frac{1}{4}x$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix}-2\\1\end{pmatrix} \Rightarrow \begin{pmatrix}1 & 2\\0 & -1\end{pmatrix}\begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}$ so $x = 0$

$\therefore y = -\frac{1}{4}x$ and $x = 0$

11) a)

$\mathbf{M} = \begin{pmatrix}0 & -\frac{2}{3}\\\frac{1}{3} & 0\end{pmatrix}$ so $\det \mathbf{M} = \frac{2}{9}$

11) b)

$$\begin{pmatrix} 0 & -\frac{2}{3} \\ \frac{1}{3} & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{2}{3}y \\ \frac{1}{3}x \end{pmatrix}$$

So that $X = -\frac{2}{3}y$ and $Y = \frac{1}{3}x \Rightarrow x = 3Y, y = -\frac{3X}{2}$

So $\frac{x^2}{9} + \frac{y^2}{4} = 1 \longmapsto \frac{(3Y)^2}{9} + \frac{\left(-\frac{3X}{2}\right)^2}{4} = 1 \Rightarrow Y^2 + \frac{9X^2}{16} = 1$

$\therefore \frac{9x^2}{16} + y^2 = 1$

11) c)

$6\pi \times \frac{2}{9} = \frac{4}{3}\pi$ units^2

12) a)

$\det \mathbf{M} = 3$

12) b)

$$\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ -x+2y \end{pmatrix}$$

So that $X = x + y$ and $Y = -x + 2y$

$X + Y = 3y \Rightarrow y = \frac{1}{3}(X + Y)$

$Y - X = y - 2x \Rightarrow Y - X = \frac{1}{3}X + \frac{1}{3}Y - 2x \Rightarrow 2x = \frac{4}{3}X - \frac{2}{3}Y \Rightarrow x = \frac{2}{3}X - \frac{1}{3}Y$

$\Rightarrow x = \frac{2}{3}X - \frac{1}{3}Y, y = \frac{1}{3}X + \frac{1}{3}Y$

So $x^2 + y^2 = 9 \longmapsto \left(\frac{2}{3}X - \frac{1}{3}Y\right)^2 + \left(\frac{1}{3}X + \frac{1}{3}Y\right)^2 = 9$

$\Rightarrow \frac{4}{9}X^2 - \frac{4}{9}XY + \frac{1}{9}Y^2 + \frac{1}{9}X^2 + \frac{2}{9}XY + \frac{1}{9}Y^2 = 9 \Rightarrow \frac{5}{9}X^2 - \frac{2}{9}XY + \frac{2}{9}Y^2 = 9$

$\therefore 5x^2 - 2xy + 2y^2 = 81$

12) c)

The original curve has area 9π

$3 \times 9\pi = 27\pi$ units^2

13) a)

$\frac{1}{2}\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}\frac{1}{2}x - \frac{1}{2}y\\ \frac{1}{2}y\end{pmatrix}$

So that $X = \frac{1}{2}x - \frac{1}{2}y$ and $Y = \frac{1}{2}y$

$X + Y = \frac{1}{2}x \Rightarrow x = 2X + 2Y$ and $y = 2Y$

So $4x^2 - 8xy + 8y^2 = 1 \longmapsto 4(2X + 2Y)^2 - 8(2X + 2Y)(2Y) + 8(2Y)^2 = 3$

$\Rightarrow 16X^2 + 32XY + 16Y^2 - 32XY - 32Y^2 + 32Y^2 = 1$

$\Rightarrow 16X^2 + 16Y^2 = 1$

$\therefore x^2 + y^2 = \frac{1}{16}$

13) b)

It is a circle so C' has area $\frac{\pi}{16}$ units^2

13) c)

$\det\begin{pmatrix}\frac{1}{2} & -\frac{1}{2}\\ 0 & \frac{1}{2}\end{pmatrix} = \frac{1}{4}$

So $\frac{1}{4}$ ×area of $C = \frac{\pi}{16}$

$\therefore$ area of C is $\frac{\pi}{4}$ units^2

14) a)

$y = \frac{1}{2}x$ and $y = -\frac{1}{2}x$

14) b)

$x^2 + 8x - y^2 = (x + 4)^2 - 16 - y^2$

So $(x + 4)^2 - y^2 = 16 \Rightarrow \frac{(x+4)^2}{16} - \frac{y^2}{16} = 1$

$\frac{x^2}{16} - \frac{y^2}{4} = 1 \longmapsto \frac{(x+4)^2}{16} - \frac{y^2}{4} = 1 \longmapsto \frac{(x+4)^2}{16} - \frac{y^2}{16} = 1$

Translation by vector $\begin{pmatrix}-4\\ 0\end{pmatrix}$ and stretch parallel to y-axis, scale factor 2 (any order)

14) c)

$y = \frac{1}{2}x \longmapsto \frac{y}{2} = \frac{1}{2}(x + 4)$

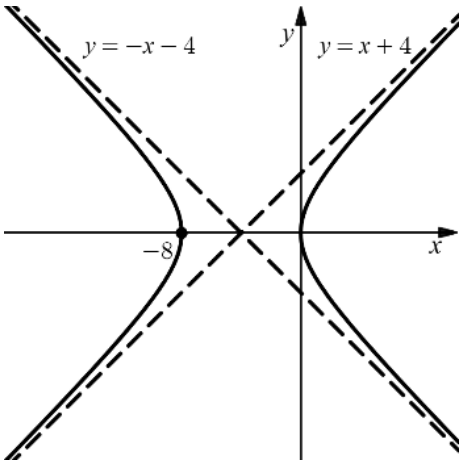
$y = -\frac{1}{2}x \longmapsto \frac{y}{2} = -\frac{1}{2}(x + 4)$

$\therefore y = x + 4$ and $y = -x - 4$

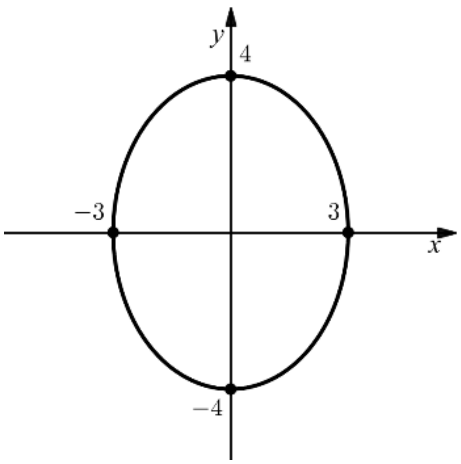
14) d)

When $x = 0$, $y = 0$ and when $y = 0$ $x = 0$ or $x = -8$

14) e)



15) a)



15) b)

$\frac{x^2}{9}+\frac{y^2}{16}=1$

Stretch along y-axis scale factor $\frac{1}{4}$:

$\frac{x^2}{9}+\frac{y^2}{16}=1 \mapsto \frac{x^2}{9}+\frac{(4y)^2}{16}=1 \Rightarrow \frac{x^2}{9}+y^2=1$

Stretch along x-axis scale factor $\frac{1}{3}$:

$\frac{x^2}{9}+y^2=1 \mapsto \frac{(3x)^2}{9}+y^2=1 \Rightarrow x^2+y^2=1$

$\therefore$ Stretch along y-axis scale factor $\frac{1}{4}$ and stretch along x-axis scale factor $\frac{1}{3}$ (any order)

15) c)

$\pi \times 1^2 \times 4 \times 3 = 12\pi$ units^2

15) d)

$\frac{(x-p)^2}{9}+\frac{(y-q)^2}{16}=1$

Touching x-axis: $q=4$

Touching y-axis: $p=3$

$\therefore p=3$ and $q=4$

15) e)

$\frac{\left(\frac{x}{2}\right)^2}{9}+\frac{y^2}{16}=1 \Rightarrow \frac{x^2}{36}+\frac{y^2}{16}=1$

$\therefore 4x^2+9y^2=144$

16) a)

When $x=0$, $y=\pm 1$ and when $y=0$, $x=\pm 4$

$\therefore (0,1), (0,-1), (4,0)$ and $(-4,0)$

16) b)

$6(x^2+4x)+y^2+2y=-19 \Rightarrow 6(x+2)^2-24+(y+1)^2-1=-19$

$\Rightarrow 6(x+2)^2+(y+1)^2=6 \Rightarrow (x+2)^2+\frac{(y+1)^2}{6}=1$

$\frac{x^2}{16}+y^2=1 \mapsto x^2+y^2=1 \mapsto x^2+\frac{y^2}{6}=1 \mapsto (x+2)^2+\frac{(y+1)^2}{6}=1$

Stretch parallel to x-axis, scale factor $\frac{1}{4}$ and stretch parallel to y-axis, scale factor $\sqrt{6}$ (any order) followed by a translation by vector $\begin{pmatrix}-2\\-1\end{pmatrix}$

16) c)

When $x=0$, y has no real solutions and when $y=0$, $x=-2\pm\sqrt{\frac{5}{6}}$

$\therefore \left(-2+\sqrt{\frac{5}{6}},0\right)$ and $\left(-2-\sqrt{\frac{5}{6}},0\right)$

16) d)

Max and mins occur on $\frac{x^2}{16}+y^2=1$ when $x=\pm4$ and $y=\pm1$

So once transformed:

$x=4 \mapsto x=1 \mapsto x=1 \mapsto x=-1$

$x=-4 \mapsto x=-1 \mapsto x=-1 \mapsto x=-3$

$y=1 \mapsto y=1 \mapsto y=\sqrt{6} \mapsto y=\sqrt{6}-1$

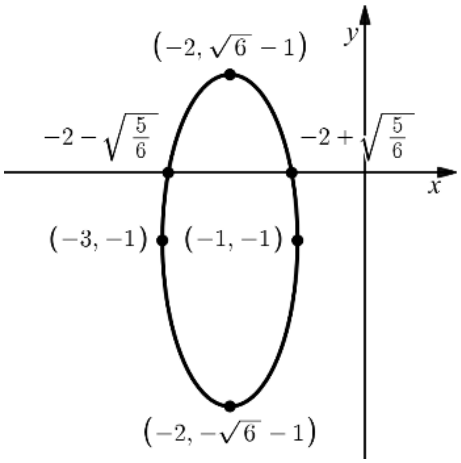
$y=-1 \mapsto y=-1 \mapsto y=-\sqrt{6} \mapsto y=-\sqrt{6}-1$

So $x_{\max}=-1$, $x_{\min}=-3$, $y_{\max}=\sqrt{6}-1$, $y_{\min}=-\sqrt{6}-1$

16) e)

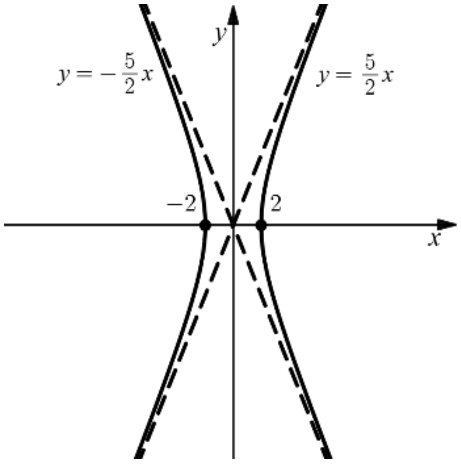
When $x = -1$, $y = -1$, when $x = -3$, $y = -1$

When $y = \sqrt{6} - 1$, $x = -2$, when $y = -\sqrt{6} - 1$, $x = -2$



17) a)

When $x = 0, y = \pm 2$



17) b)

Translation by vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$: $(x, y) \mapsto (x - k, y)$

$\frac{x^2}{4} - \frac{y^2}{25} = 1 \mapsto \frac{(x-k)^2}{4} - \frac{y^2}{25} = 1$

Reflection in the line $y = -x$: $(x, y) \mapsto (-y, -x)$

$\frac{(x-k)^2}{4} - \frac{y^2}{25} = 1 \mapsto \frac{(-y-k)^2}{4} - \frac{x^2}{25} = 1$

This passes through the origin, $(0, 0)$:

$\frac{(0-k)^2}{4} - \frac{0}{25} = 1 \Rightarrow k^2 = 4 \therefore k = \pm 2$

Since k is positive, $k = 2$

17) c)

Translation by vector $\begin{pmatrix}2\\0\end{pmatrix}$: $(x,y)\longmapsto(x-2,y)$

$y=\pm\frac{5}{2}x\longmapsto y=\pm\frac{5}{2}(x-2)$

Reflection in the line $y=-x$: $(x,y)\longmapsto(-y,-x)$

$y=\pm\frac{5}{2}(x-2)\longmapsto -x=\pm\frac{5}{2}(-y-2)$

So $-x=\frac{5}{2}(-y-2)\Rightarrow y=\frac{2}{5}x-2$ or $-x=-\frac{5}{2}(-y-2)\Rightarrow y=-\frac{2}{5}x-2$

18) a)

Translation by vector $\begin{pmatrix}2\\-1\end{pmatrix}$: $(x,y)\longmapsto(x-2,y+1)$

$\frac{x^2}{4}-\frac{y^2}{3}=1\longmapsto\frac{(x-2)^2}{4}-\frac{(y+1)^2}{3}=1$

a rotation anti-clockwise by $\frac{\pi}{2}$: $(x,y)\longmapsto(y,-x)$

$\frac{(x-2)^2}{4}-\frac{(y+1)^2}{3}=1\longmapsto\frac{(y-2)^2}{4}-\frac{(-x+1)^2}{3}=1$

a stretch in the x-direction by factor 3: $(x,y)\longmapsto\left(\frac{x}{3},y\right)$

$\frac{(y-2)^2}{4}-\frac{(-x+1)^2}{3}=1\longmapsto\frac{(y-2)^2}{4}-\frac{\left(-\frac{x}{3}+1\right)^2}{3}=1\Rightarrow\frac{(y-2)^2}{4}-\frac{(x-3)^2}{27}=1$

So that $\frac{(y-2)^2}{4}-\frac{(x-3)^2}{27}=1\Rightarrow 27(y-2)^2-4(x-3)^2=108$

$\Rightarrow 27(y^2-4y+4)-4(x^2-6x+9)=108$

$\Rightarrow 27y^2-108y+108-4x^2+24x-36=108$

$\Rightarrow 27y^2-108y-4x^2+24x=36\Rightarrow 4x^2-27y^2-24x+108y+36=0$

18) b)

For $\frac{x^2}{4} - \frac{y^2}{3} = 1$, $(-2, 0)$ and $(2, 0)$

$(-2, 0) \longmapsto (0, -1) \longmapsto (1, 0) \longmapsto (3, 0)$

$(2, 0) \longmapsto (4, -1) \longmapsto (1, 4) \longmapsto (3, 4)$

So $k < 0$ or $k > 4$

$\therefore \{k: k \in \mathbb{R}, k < 0\} \cup \{k: k \in \mathbb{R}, k > 4\}$

Alternatively:

When $y = k$,

$4x^2 - 27k^2 + 108k - 24x + 36 = 0 \Rightarrow 4x^2 - 24x + 36 + 108k - 27k^2 = 0$

Two points of intersection: $b^2 - 4ac > 0$

$(-24)^2 - 4(4)(36 + 108k - 27k^2) > 0$

$432k^2 - 1728k > 0$

$432k(k - 4) > 0$

So $k < 0$ or $k > 4$

$\therefore \{k: k \in \mathbb{R}, k < 0\} \cup \{k: k \in \mathbb{R}, k > 4\}$

19) a)

$4x - 3y = 11 \Rightarrow y = \frac{4}{3}x - \frac{11}{3}$

$\frac{x^2}{9} - \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{9} - \frac{\left(\frac{4}{3}x - \frac{11}{3}\right)^2}{16} = 1 \Rightarrow \frac{x^2}{9} - \frac{(4x-11)^2}{144} = 1 \Rightarrow 16x^2 - (4x - 11)^2 = 144$

$\Rightarrow 16x^2 - 16x^2 + 88x - 121 = 144 \Rightarrow 88x = 265 \Rightarrow x = \frac{265}{88}$

One solution and so only one point of intersection.

19) b)

$y = \frac{4}{3}x$ is translated: $(y + 1) = \frac{4}{3}(x - a)$

$3y + 3 = 4x - 4a \Rightarrow 4a + 3 = 4x - 3y$

But since $4x - 3y = 11$ is now an asymptote, $4a + 3 = 11$

$\therefore a = 2$

20) a)

Consider the points $(5,0)$, $(0,2)$, $(-5,0)$ and $(0,-2)$

And apply: reflection across the line $y=x$, Translation by vector $\begin{pmatrix}a\\b\end{pmatrix}$ and enlargement of scale factor 2

$(-5,0)$ will be transformed to the x-axis

$(0,-2)$ will be transformed to the y-axis

$(-5,0) \mapsto (0,-5) \mapsto (a,-5+b) \mapsto (2a,-10+2b)$

$\therefore -10+2b=0 \Rightarrow b=5$

$(0,-2) \mapsto (-2,0) \mapsto (a-2,b) \mapsto (2a-4,2b)$

$\therefore 2a-4=0 \Rightarrow a=2$

$\therefore\ a=2$ and $b=5$

20) b)

$\frac{x^2}{25}+\frac{y^2}{4}=1 \mapsto \frac{x^2}{4}+\frac{y^2}{25}=1 \mapsto \frac{(x-2)^2}{4}+\frac{(y-5)^2}{25}=1 \mapsto \frac{\left(\frac{x}{2}-2\right)^2}{4}+\frac{\left(\frac{y}{2}-5\right)^2}{25}=1$

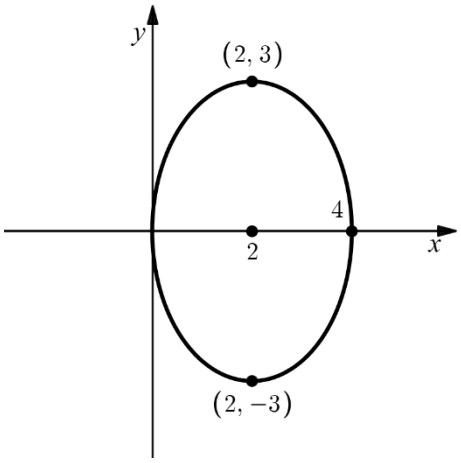
$\frac{\left(\frac{x}{2}-\frac{4}{2}\right)^2}{4}+\frac{\left(\frac{y}{2}-\frac{10}{2}\right)^2}{25}=1$

$\therefore \frac{(x-4)^2}{16}+\frac{(y-10)^2}{100}=1$

21) a)

Ellipse

21) b)



The translation must have negative values for both a and b so that the curve is translated in the left and downwards direction.

Given we know after being reflected in the y-axis, it will touch the curve C_1 just once, the max point of C_2 must be $(2, -3)$

This point has been reflected from the point $(-2, -3)$

Therefore the maximum of C_1 is translated from $(2, 3)$ to $(-2, -3)$ which is translation $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$

$\therefore a = -4,\ b = -6$

21) c)

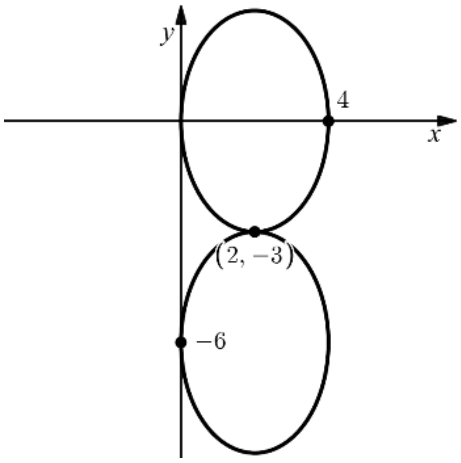
$\frac{(x-2)^2}{4} + \frac{y^2}{9} = 1 \mapsto \frac{(x+2)^2}{4} + \frac{(y+6)^2}{9} = 1 \mapsto \frac{(-x+2)^2}{4} + \frac{(y+6)^2}{9} = 1$

$\therefore \frac{(x-2)^2}{4} + \frac{(y+6)^2}{9} = 1$

21) d)

$(2, -3)$

21) e)



21) f)

$x = 0$, $x = 4$ and $y = -3$

22) a)

$(2, 0)$, $\left(0, \sqrt{2}\right)$, $(-2, 0)$ and $\left(0, -\sqrt{2}\right)$

22) b)

Translation vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$: $(x, y) \longmapsto (x - k, y)$

$\frac{x^2}{4} + \frac{y^2}{2} = 1 \longmapsto \frac{(x-k)^2}{4} + \frac{y^2}{2} = 1$

Enlargement of scale factor $\sqrt{3}$: $(x, y) \longmapsto \left(\frac{x}{\sqrt{3}}, \frac{y}{\sqrt{3}}\right)$

$\frac{(x-k)^2}{4} + \frac{y^2}{2} = 1 \longmapsto \frac{\left(\frac{x}{\sqrt{3}} - k\right)^2}{4} + \frac{\left(\frac{y}{\sqrt{3}}\right)^2}{2} = 1$

$\therefore \frac{\left(x - \sqrt{3}k\right)^2}{12} + \frac{y^2}{6} = 1$

22) c)

$\frac{\left(x - \sqrt{3}k\right)^2}{12} + \frac{y^2}{6} = 1$ when $y = -\frac{1}{2}x + 5$:

$\frac{\left(x - \sqrt{3}k\right)^2}{12} + \frac{\left(-\frac{1}{2}x + 5\right)^2}{6} = 1 \Rightarrow \left(x - \sqrt{3}k\right)^2 + 2\left(-\frac{1}{2}x + 5\right)^2 = 12$

$\Rightarrow x^2 - 2\sqrt{3}kx + 3k^2 + \frac{1}{2}x^2 - 10x + 50 = 12$

$\Rightarrow 2x^2 - 4\sqrt{3}kx + 6k^2 + x^2 - 20x + 76 = 0$

$\therefore 3x^2 - \left(4\sqrt{3}k + 20\right)x + 6k^2 + 76 = 0$ as req.

22) d)

One (repeated) solution: $b^2 - 4ac = 0$

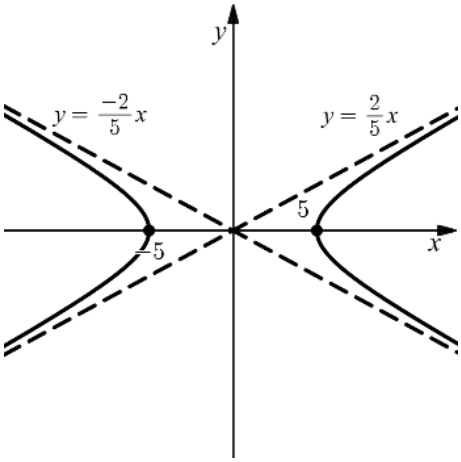
$\left(4\sqrt{3}k + 20\right)^2 - 4(3)(6k^2 + 76) = 0$

$\Rightarrow 48k^2 + 160\sqrt{3}k + 400 - 72k^2 - 912 = 0$

$\Rightarrow 24k^2 - 160\sqrt{3}k + 512 = 0$

$\therefore k = \frac{4\sqrt{3}}{3}$ or $\frac{16\sqrt{3}}{3}$

23) a)



23) b)

$\frac{x^2}{25} - \frac{(x+k)^2}{4} = 1 \Rightarrow 4x^2 - 25(x+k)^2 = 100$

$\Rightarrow 4x^2 - 25x^2 - 50kx - 25k^2 = 100$

$\Rightarrow 21x^2 + 50kx + 100 + 25k^2 = 0$

Two distinct solutions, when $b^2 - 4ac > 0$:

$(50k)^2 - 4(21)(100 + 25k^2) > 0 \Rightarrow 2500k^2 - 2100k^2 - 8400 > 0$

$\Rightarrow 400k^2 - 8400 > 0$

$\Rightarrow k^2 - 21 > 0$

$\therefore k > \sqrt{21}$ and $k < -\sqrt{21}$ as req.

23) c)

$\frac{(x-a)^2}{25} - \frac{(y-b)^2}{4} = 1$

$\Rightarrow 4(x-a)^2 - 25(y-b)^2 = 100$

$\Rightarrow 4x^2 - 8ax + 4a^2 - 25y^2 + 50by - 25b^2 - 100 = 0$

So $-8a = -32 \Rightarrow a = 4$, $50b = 50 \Rightarrow b = 1$

Substitute in:

$4x^2 - 32x + 64 - 25y^2 + 50y - 25 - 100 = c \Rightarrow c = 61$

$\therefore a = 4, b = 1$ and $c = 61$

23) d)

Let $y = x + p$ be a line parallel to $y = x$

$4x^2 - 25(x+p)^2 - 32x + 50(x+p) = 61$

$\Rightarrow 4x^2 - 25p^2 - 50px - 25x^2 - 32x + 50x + 50p = 61$

$\Rightarrow 21x^2 + (50p - 18)x + 25p^2 - 50p + 61 = 0$

Tangent, when $b^2 - 4ac = 0$:

$(50p - 18)^2 - 4(21)(25p^2 - 50p + 61) = 0$

$400p^2 + 2400p - 4800 = 0$

$\therefore p = -3 \pm \sqrt{21}$

Which gives $y = x - 3 + \sqrt{21}$ and $y = x - 3 - \sqrt{21}$

24) a)

$\frac{x^2}{4} + \frac{y^2}{100} = 1 \Rightarrow 25x^2 + (2x + k)^2 = 100$

$\Rightarrow 25x^2 + 4x^2 + 4kx + k^2 = 100$

$\Rightarrow 29x^2 + 4kx + k^2 - 100 = 0$

Two distinct solutions, when $b^2 - 4ac > 0$:

$(4k)^2 - 4(29)(k^2 - 100) > 0 \Rightarrow 16k^2 - 116k^2 + 11600 > 0$

$\Rightarrow -100k^2 + 11600 > 0$

$\Rightarrow k^2 - 116 < 0$

$\therefore -\sqrt{116} < k < \sqrt{116}$ as req.

24) b)

$\frac{(x-a)^2}{4}+\frac{(y-b)^2}{100}=1$

$\Rightarrow 25(x-a)^2+(y-b)^2=100$

$\Rightarrow 25x^2-50ax+25a^2+y^2-2by+b^2-100=0$

So $-50a=-150 \Rightarrow a=3$, $-2b=10 \Rightarrow b=-5$

Substitute in:

$25x^2-150x+225+y^2+10y+25-100=c \Rightarrow c=-150$

$\therefore a=3, b=-5$ and $c=-150$

24) c)

Let $y=2x+p$ be a line parallel to $y=2x$

$25x^2+(2x+p)^2-150x+10(2x+p)=-150$

$\Rightarrow 25x^2+4x^2+4px+p^2-150x+20x+10p+150=0$

$\Rightarrow 29x^2+(4p-130)x+p^2+10p+150=0$

Tangent, when $b^2-4ac=0$:

$(4p-130)^2-4(29)(p^2+10p+150)=0$

$p^2+22p+5=0$

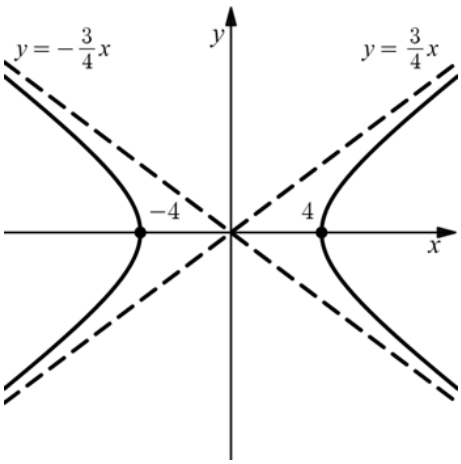
$\therefore p=-11\pm 2\sqrt{29}$

Which gives $y=2x-11+2\sqrt{29}$ and $y=2x-11-2\sqrt{29}$

25) a)

Hyperbola

25) b)



25) c)

The asymptotes $y = 3x - k$ and $y = k - 3x$ intersect at centre of the curve C_2:

$3x - k = k - 3x \Rightarrow 6x = 2k \Rightarrow x = \frac{k}{3}$

When $x = \frac{k}{3}$, $y = 3 \times \frac{k}{3} - k = 0 \therefore \left(\frac{k}{3}, 0\right)$

When $k = 6$, asymptotes of C_2 pass through $(2, 0)$ so the translation is given by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$\frac{x^2}{16} - \frac{y^2}{9} = 1 \mapsto \frac{(x-2)^2}{16} - \frac{y^2}{9} = 1$

Consider the original asymptote: $y = \frac{3}{4}x$

$y = \frac{3}{4}x \mapsto y = \frac{3}{4}(x - 2) = \frac{3}{4}x - \frac{6}{4}$

So stretch is a stretch along the y-axis scale factor 4

$\frac{(x-2)^2}{16} - \frac{\left(\frac{y}{4}\right)^2}{9} = 1 \mapsto \frac{(x-2)^2}{16} - \frac{y^2}{144} = 1$

$\therefore \frac{(x-2)^2}{16} - \frac{y^2}{144} = 1$

25) d)

As k is positive, it only gives a translation in the positive x-direction by vector $\begin{pmatrix} \frac{k}{3} \\ 0 \end{pmatrix}$:

$\frac{\left(x-\frac{k}{3}\right)^2}{16} - \frac{y^2}{144} = 1$

Three intersections are where they are tangent at one point:

When $y = 0$:

$\frac{x^2}{16} - \frac{0^2}{9} = 1 \Rightarrow x = \pm 4$

$\frac{\left(x-\frac{k}{3}\right)^2}{16} - \frac{0^2}{144} = 1 \Rightarrow x - \frac{k}{3} = \pm 4 \Rightarrow x = \frac{k}{3} \pm 4$

So $4 = \frac{k}{3} - 4 \therefore k = 24$